

GAIRA

Cross-Modal Transfer — the full report

What survives when a pure-compound spectrum moves from Raman to Ag-SERS, measured as a five-level Representation Hierarchy — latent fingerprint, MSS motif, biochemical theme, perturbation, and matrix robustness — with honest null controls throughout.

Frozen atlas fingerprint 09ed804a40836f4a05a91ba10900cded

51 matched pure analytes · Raman reference vs pure Ag-SERS · SERS validates the model, never trains it

A document version of Foundation Explorer V3, incorporating the V1 (latent) and V2 (theme) analyses. Every figure is reproduced from committed artifacts; nothing is retrained.

Executive summary

There is no single number for “how well does Raman transfer to Ag-SERS.” Transfer is a **hierarchy of representations**, and the honest question is how far up the hierarchy the agreement survives — because each level is progressively more abstract and less governed by silver’s surface physics.

Read top to bottom, two things happen at once. The **raw** agreement rises (latent 0.4249 → MSS 0.7405 → theme 0.9181), which looks like “more biochemistry survives the higher you go.” But the **identity-specific** agreement falls at the top (theme identity 0.1114, argmax 35%). Both are true, and holding them together is the whole point.

The central caution. Raw theme cosine (0.92) and raw Spearman rank ρ (0.87) are spectacular but **baseline-inflated**: every analyte’s theme composition shares the same dominant background, so any two analytes — even unrelated ones — already agree at ≈ 0.9 before preservation is considered. Once the baseline is removed, identity-specific preservation is selective, concentrated in the strong silver adsorbers.

The useful minority. A real subset — adenine foremost — redistributes its latent fingerprint yet keeps a **dose-responsive** biochemical theme. Adenine is weak at Level 1 (0.36) yet responds to concentration along a saturating Langmuir law ($\rho=0.996$). A single cosine would call it a failure; the hierarchy shows it is one of GAIRA’s best-validated analytes.

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1 · The Representation Hierarchy

Each level below is more abstract, and less governed by adsorption physics, than the one above it. GAIRA reports the whole ladder, never a single collapsed “preservation score.”

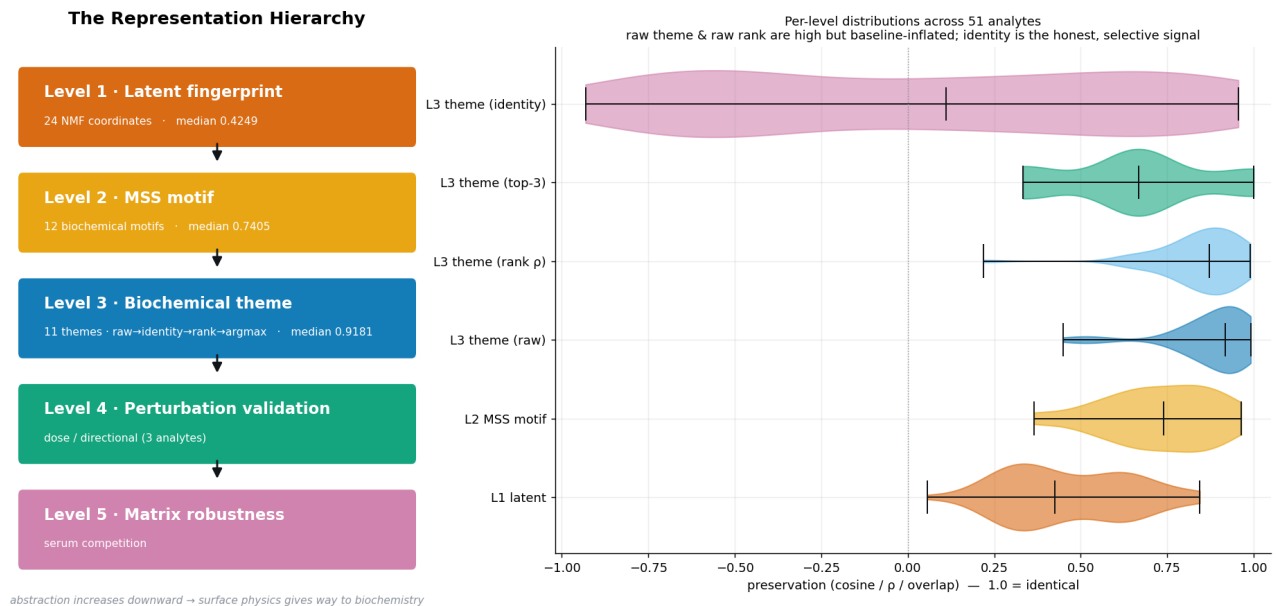


Figure 1. The conceptual ladder (left) and the per-level distributions across 51 analytes (right). Raw theme and raw rank cluster high but are baseline-inflated; the identity metric spreads wide — the honest, selective signal.

Per-level summary

Level	Metric	median	mean	std	min	max
1 · Latent fingerprint	component cosine	0.425	0.464	0.182	0.056	0.844
2 · MSS motif	mss cosine	0.741	0.725	0.156	0.365	0.964
3 · Theme (raw)	theme cosine raw	0.918	0.873	0.124	0.450	0.991
3 · Theme (identity)	baseline-subtracted cosine	0.111	0.047	0.612	-0.932	0.956
3 · Theme (rank ρ)	Spearman rank	0.873	0.841	0.129	0.218	0.991
3 · Theme (top-3)	top-3 overlap	0.667	0.654	0.231	0.333	1.000

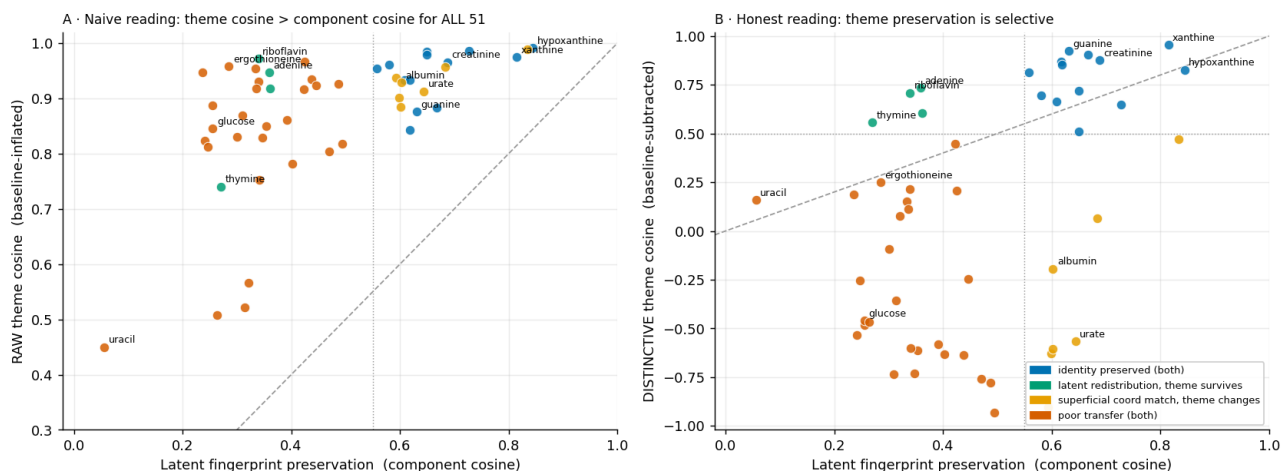
Read the hierarchy top-to-bottom: raw agreement rises, identity-specific agreement falls. The job is not to maximise agreement but to separate surface physics from biochemical meaning.

2 · The metric problem

The original pure-Ag-SERS stage reports one number per analyte — the cosine between the 24 NMF coordinates — and calls it *recoverability*. That number is correct and is kept (it becomes Level 1). But it measures only whether two spectra land on the same point of the latent manifold; it says nothing about whether the biochemical interpretation survives, whether a perturbation would register, or whether the analyte is recoverable in serum. Those are four different questions.

At the theme level the raw cosine is 0.9181 — far above the latent 0.4249, and it exceeds the latent cosine for all 51 analytes. Read naively this “proves” the theme almost always survives. It does not: the composition of every analyte is dominated by the same high-share background themes, so two unrelated analytes already sit at ≈ 0.9 . The fix is to subtract the shared baseline and measure each analyte’s distinctive deviation against a null.

Cross-modal transfer: latent fingerprint vs biochemical theme (Raman → Ag-SERS, 51 pure analytes)



Left: raw theme cosine sits above the diagonal for every analyte — but this is compositional-baseline inflation, not preservation. Right: baseline-subtracted, only strong chemisorbers keep an identity-specific theme.

Figure 2. Naive vs honest reading (from the V2 analysis). Left: raw theme cosine sits above the diagonal for every analyte — baseline inflation. Right: baseline-subtracted, only the strong chemisorbers keep an identity-specific theme.

3 · Metric specification

The theme level is a **ladder of strictness** — raw → identity → rank → top-k → argmax. Each answers a different question; the raw ones are baseline-inflated and are never read alone. Symbols: z = 24 NMF coordinates, m = 12 MSS motifs, t = 11 theme shares, subscripts R (Raman) / S (Ag-SERS); b = the shared baseline (mean Raman theme composition over all analytes).

Metric	Equation	Result	Purpose / limitation
L1 Latent fingerprint	$\cos(z_R, z_S)$	median 0.4249	Do the coordinates line up? Adsorption-dominated — a low value is surface physics, not a representation error.
L2 MSS motif	$\cos(m_R, m_S)$	median 0.7405	Mid-level structure; more robust than coordinates, still surface-sensitive.
L3a Theme raw	$\cos(t_R, t_S)$	median 0.9181	Gross composition similarity. CONTAINS BASELINE INFLATION — never a stand-alone measure.
L3b Theme identity	$\cos(t_R - b, t_S - b)$ vs null	median 0.1114	Baseline-subtracted, self-referenced against other analytes. The honest, selective theme signal.
L4 Theme rank (NEW)	Spearman $\rho(\text{rank } t_R, \text{rank } t_S)$	median 0.8727	Ordering of all 11 themes. Robust to argmax instability — but ALSO baseline-inflated (separation 0.0102).
L5 Top-k overlap (NEW)	$ \text{topk}(t_R) \cap \text{topk}(t_S) /k$	top-3 0.667	Do the leading themes stay in the top set? Interpretable middle-ground; avoids argmax knife-edge.
L6 Argmax	$\text{argmax}(t_R) == \text{argmax}(t_S)$	35%	Strict, unstable single-dominant test. On Ag, 50/51 become purine — so agreement mostly means ‘already purine in Raman’.

Reading rule adopted across GAIRA: never quote a raw metric (L3a cosine or L4 rank ρ) without its null / separation; report the hierarchy, not a single score; use physics-aware language (latent redistribution, adsorption-driven observation bias, identity-specific preservation, functional perturbation validation), never “theme preserved / failed.”

LEVEL 1**Latent fingerprint preservation**

Cosine between the 24 NMF coordinates — the finest-grained view, dominated by adsorption physics. Median **0.4249** (mean 0.4638, range [0.0562, 0.8443]). Strong Ag chemisorbers (oxopurines) score high; weak physisorbers (pyrimidines, sugars, small amino acids) low. A low value here is a surface effect, not a failure of the frozen representation — this is the original V1 “coordinate cosine,” reframed as Level 1 and kept verbatim.

LEVEL 2**MSS motif preservation**

Cosine over the 12 biochemical MSS motif activations — median **0.7405**, sitting between the latent coordinates (0.42) and the raw theme (0.92). Motif structure is more robust than exact coordinates but still adsorption-sensitive; it is also the layer at which perturbations localise (the uricase case, §11).

LEVEL 3

Biochemical theme — raw & identity-specific

The interpretation layer, measured as a ladder. The two cosines first: **raw** (median 0.9181, baseline-inflated) and **identity-specific** (median 0.1114, baseline-subtracted vs a null over the other analytes). Identity-specific preservation is real but **selective** — concentrated in the strong silver adsorbers (oxopurines, cofactors, creatinine/urea, PEP/citrate).

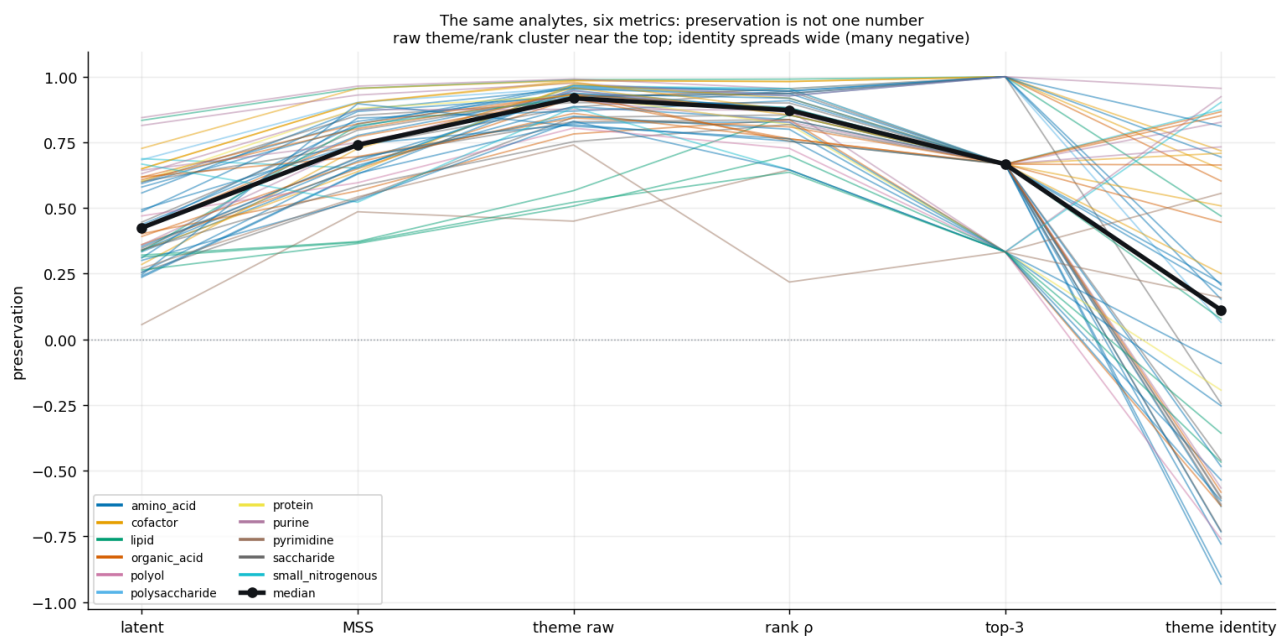


Figure 3. The same 51 analytes across six metrics: preservation is not one number. Raw theme and raw rank cluster near the top; the identity metric spreads wide, many negative.

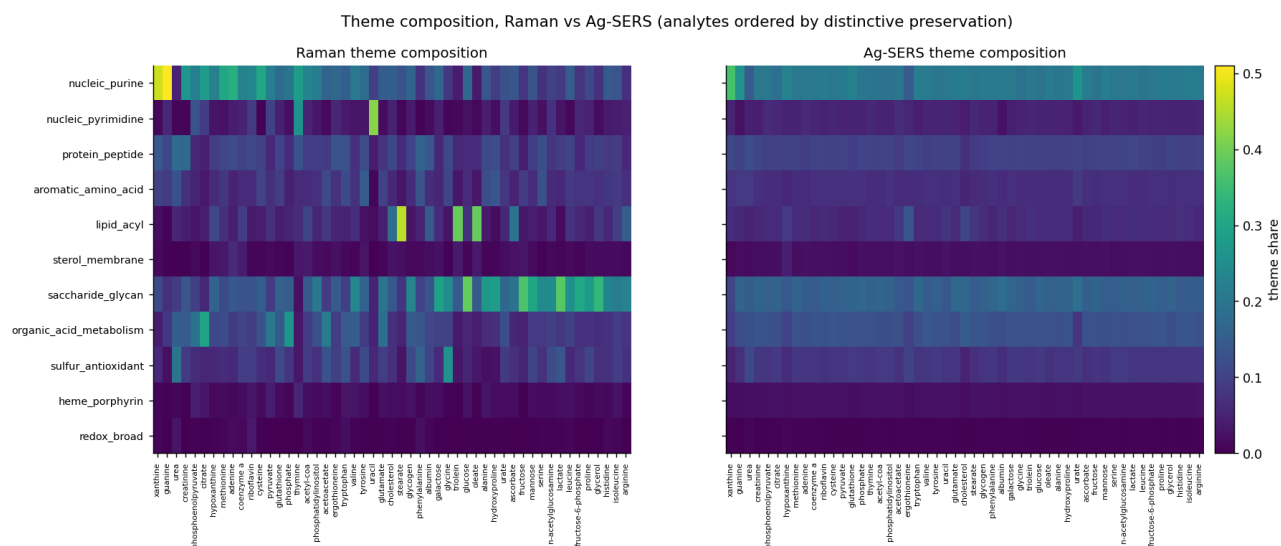


Figure 4. Theme composition, Raman (left) vs Ag-SERS (right), analytes ordered by distinctive preservation. The rich, analyte-specific Raman structure is homogenised on silver toward a purine-dominated profile.

Theme RANK preservation (Spearman) — new

Instead of argmax or magnitude cosine, compute Spearman ρ between the Raman and Ag-SERS theme *rankings*. This uses the full ordering, including minor themes, and is robust to argmax instability — likely the most representative single view. Raw ρ median **0.8727**.

Tested, not assumed. Raw rank ρ is **also baseline-inflated**: its null (rank of one analyte's Raman vs another's Ag-SERS) is ≈ 0.85 , so the identity-specific rank separation is only **0.0102** (positive for 34/51 — a slim edge over cosine's 28/51). Rank ordering carries marginally more identity signal than magnitude, and no more.

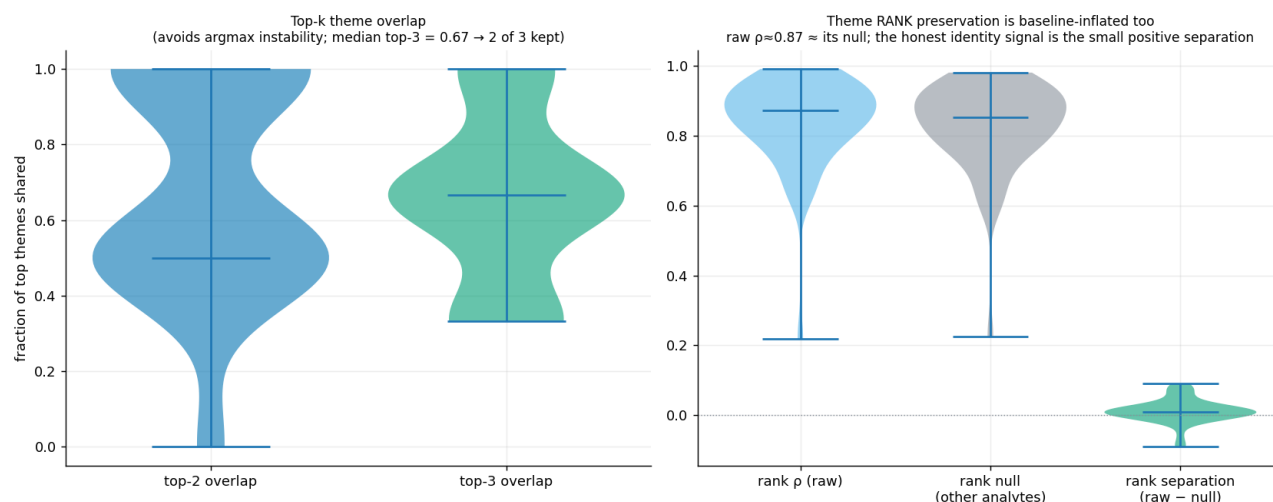


Figure 5. Right: raw rank $\rho \approx$ its null; the honest signal is the small positive separation. Left: top-k overlap (next section).

LEVEL 3

Top-k theme overlap — new

If purine, protein and organic remain within the top three themes, that is partial preservation even if the single argmax flips. Top-k overlap avoids argmax instability and is less baseline-inflated than cosine/rank because it looks only at the leading (more variable) themes. Median top-2 **0.5**, top-3 **0.667** — typically two of three leading themes retained. This is the interpretable middle-ground of the theme ladder.

LEVEL 3

Dominant-theme agreement (argmax)

Whether the single most-active theme survives — agreement **35%**. Intentionally the strictest test, and the least stable: on Ag-SERS 50/51 analytes become nucleic_purine-dominant, so all 18 agreements are analytes that were *already* purine-dominant in Raman. Argmax is a strict corner case, never the headline — which is exactly why GAIRA reports the softer rank and top-k metrics alongside it.

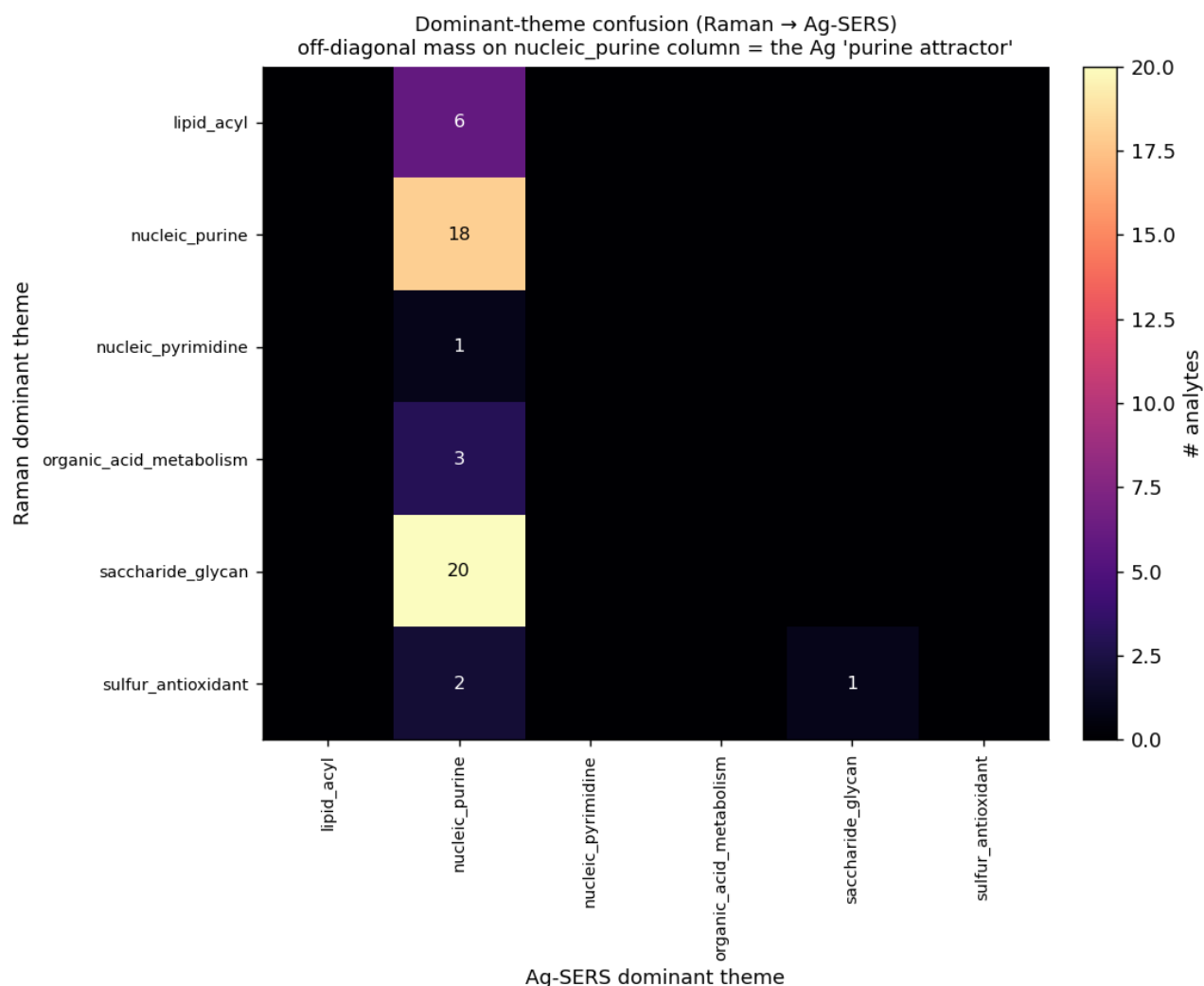


Figure 6. Dominant-theme confusion (Raman → Ag-SERS): nearly all off-diagonal mass is the nucleic_purine column — the silver purine attractor.

10 · The purine attractor, quantified

Silver colloid binds N-heterocycles strongly, so oxopurine-like signal dominates the SERS of weak adsorbers. This is the mechanism behind the low argmax agreement — and V3 quantifies it per analyte.

Δ Purine = purine share in Ag-SERS minus in Raman. It increases for **36/51** analytes (median 0.0577): non-purines gain purine share on silver, while already-purine-rich analytes lose it (regression toward the attractor). Crucially, **Δ Purine anti-correlates with the latent fingerprint ($r = -0.38$, $p = 0.006$)** — the weaker the adsorption fidelity, the harder the analyte is pulled into the attractor.

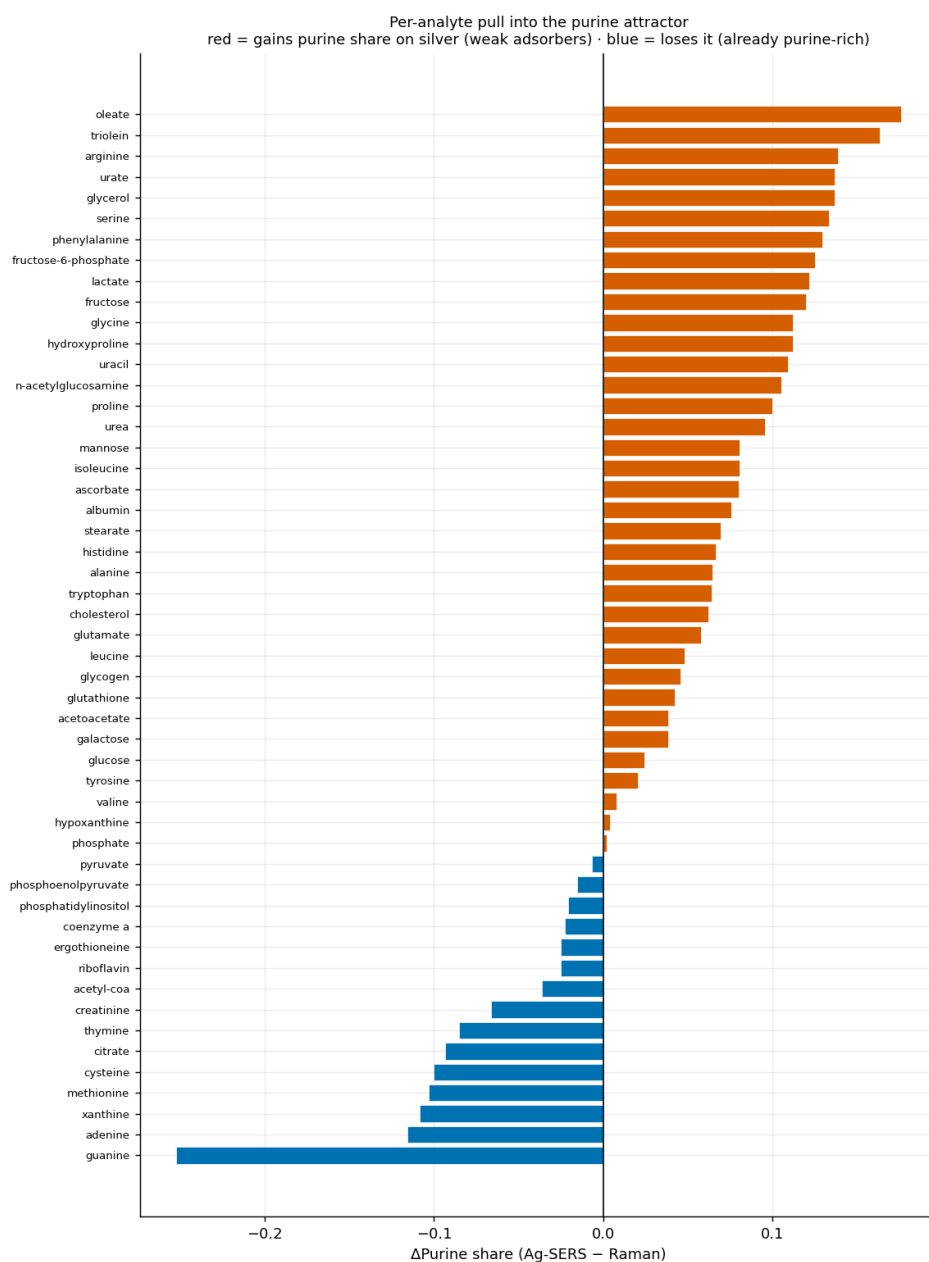


Figure 7. Per-analyte Δ Purine. Red = gains purine share on silver (weak adsorbers); blue = loses it (already purine-rich, e.g. guanine -0.25).

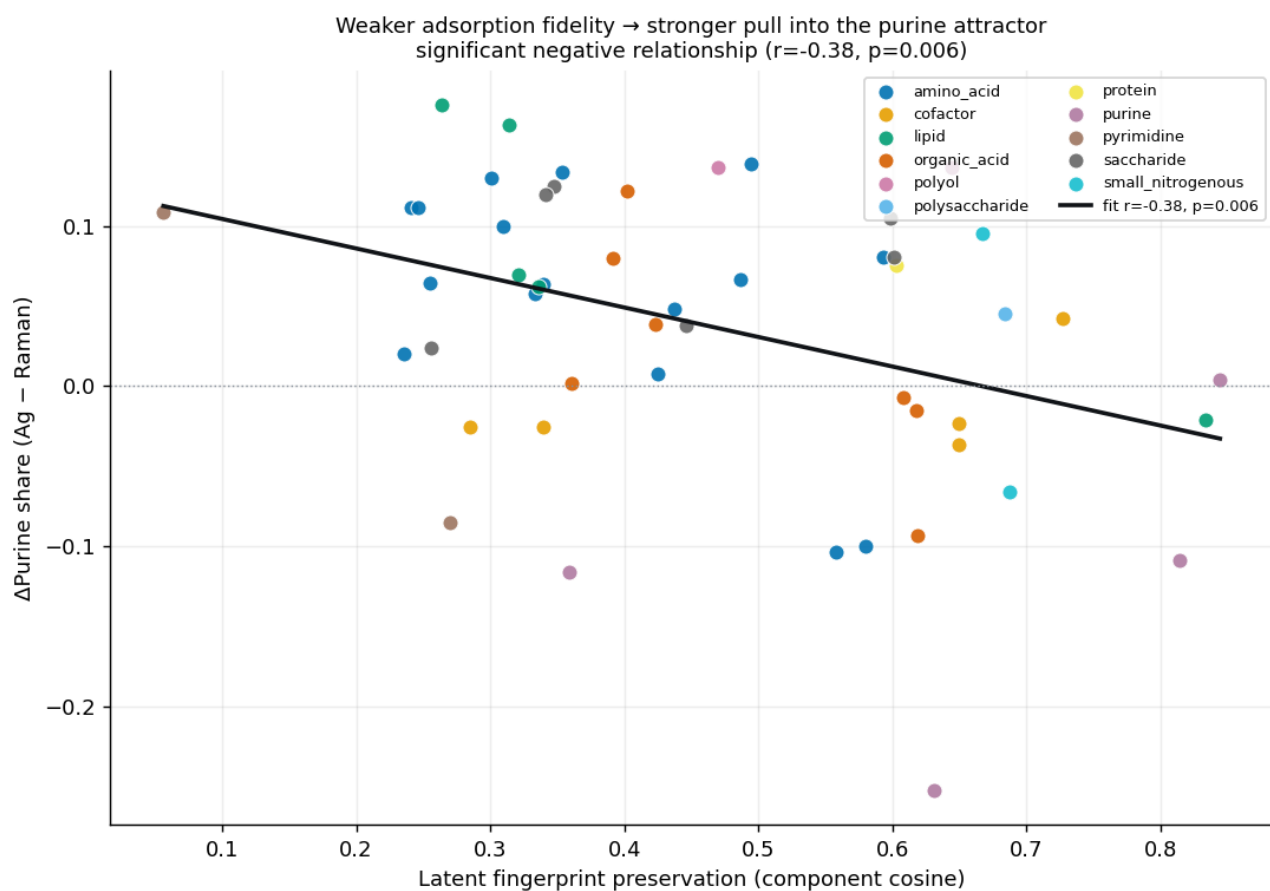


Figure 8. Significant negative relationship ($r=-0.38$, $p=0.006$): weaker adsorption fidelity → stronger attractor pull.

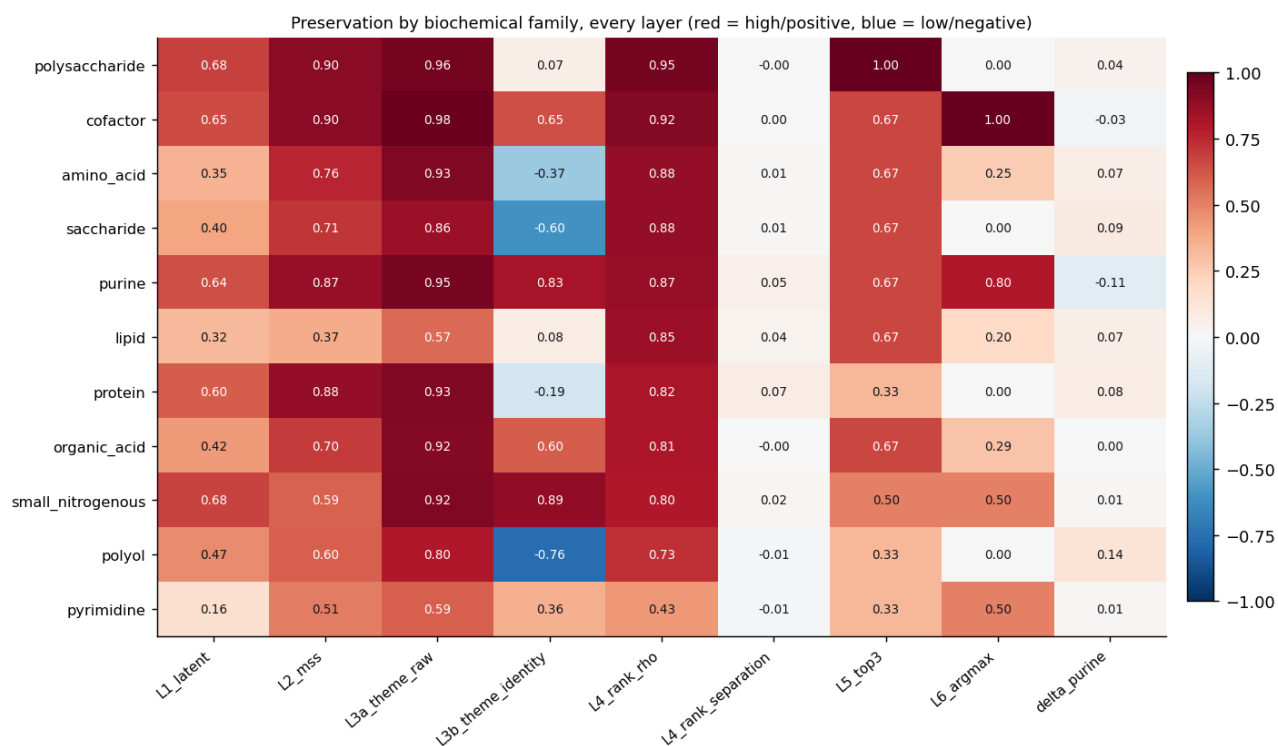


Figure 9. Preservation by biochemical family across every layer. Identity-specific preservation (column L3b) is positive for purines, cofactors, organic acids and small-nitrogenous; negative (scrambled) for amino acids, sugars, polyol.

This is adsorption-driven observation bias — a property of the silver surface and a target for a future observation model, not a defect of the frozen representation, which honestly flags the modality gap (OOD 0.05 → 0.16).

LEVEL 4

Perturbation validation

A controlled change that still moves the correct theme is stronger evidence than any static similarity — it shows the abstraction is **functional**. It exists for **exactly three analytes**; every other analyte is “not measured,” never implied.

Analyte	Type	Target	Key finding
adenine	concentration dose-response	nucleic_purine	purine theme rises monotonically and saturates (Langmuir) — a reproducible dose-response
ergothioneine	concentration dose-response	sulfur_antioxidant	sulfur theme rises monotonically and saturates — a reproducible dose-response
urate	directional depletion (uricase)	nucleic_purine (oxopurine motif)	enzymatic urate removal drops the oxopurine-carbonyl MOTIF sharply (theme layer is diffuse); validates perturbation DIRECTION, not a dose score

Level 4 — Perturbation validation exists for EXACTLY three analytes
dynamic response is stronger evidence than any static similarity — but it is rare

analyte	perturbation type	target	key metric
adenine	concentration dose-response	nucleic_purine	$\rho=0.996$, Langmuir $K=0.893 \mu\text{M}$
ergothioneine	concentration dose-response	sulfur_antioxidant	$\rho=0.927$, saturating $K=1.521 \mu\text{M}$
uricase (urate)	directional depletion (NOT a dose)	oxopurine motif	$\Delta\text{oxopurine motif} = -0.0602$

Figure 10. The three validated perturbations. Adenine and ergothioneine are saturating dose-responses; uricase is a directional depletion (not a dose).

Adenine — weak at Level 1 (0.36) — drives the purine theme monotonically ($\rho=0.996$) along a saturating Langmuir law ($K=0.89 \mu\text{M}$): the abstraction tracks concentration, not coincidence. Dynamic response is the strongest rung in the ladder — but only 3 of 51 analytes provide it, and GAIRA never extrapolates it.

LEVEL 5

Matrix robustness

Does pure-Ag transfer *predict* serum recoverability across all 51 analytes? Regression says only weakly: latent fingerprint → serum spike displacement gives **$r=0.167$, $R^2=0.0279$, $p=0.241$ (not significant)**; the Spearman ρ is 0.073. The top oxopurines survive both the modality and the matrix gap, but there is no tight per-analyte law — serum adds substantial matrix-specific competition beyond pure adsorption strength.

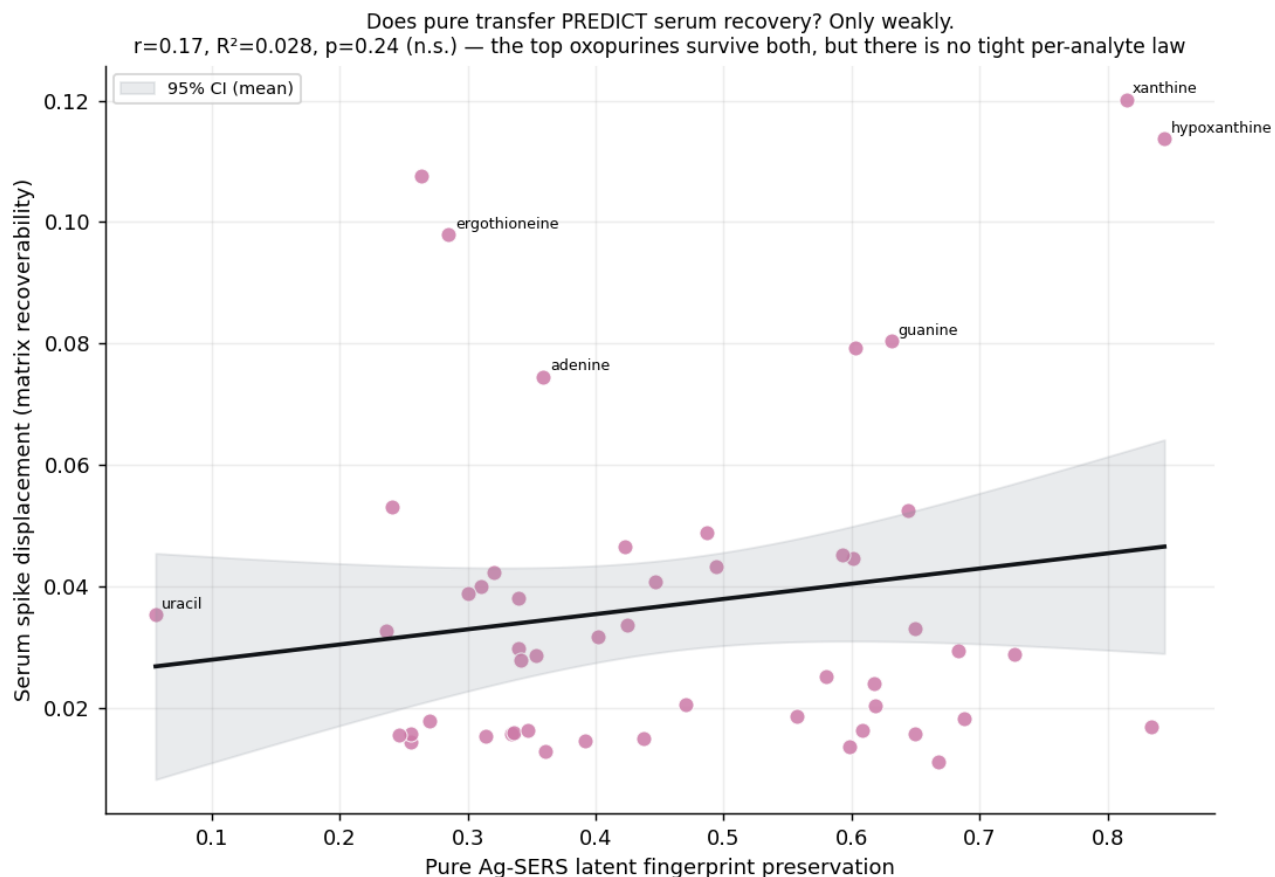


Figure 11. Pure transfer vs serum spike displacement with 95% CI. The slope is shallow and its CI spans zero — a categorical top-set agreement, not a quantitative law. This is an honest downgrade of the earlier categorical claim.

Serum data validate **matrix** recoverability, not pure-analyte theme preservation. The two are kept separate throughout the framework.

13 · Cross-modal transfer map & 14 · Per-analyte assessment

Each analyte carries nine layers: latent · MSS · theme cosine · rank p · top-3 · argmax · family · interpretation · limitations. Four exemplars span the behaviour space.

adenine — purine

L1 latent 0.3592 · L2 MSS 0.7405 · theme raw 0.9466 · theme identity 0.733 · rank p 0.8636 (sep 0.0331) · top-3 0.667 · argmax nucleic_purine→nucleic_purine · Δ Purine -0.1157

Interpretation. Latent redistribution with retained biochemical abstraction: the 24-coordinate fingerprint is reshaped by adsorption, yet the higher-level theme identity survives. functional perturbation validation — a 14-point concentration series drives the nucleic_purine theme monotonically ($p=0.996$) along a saturating Langmuir law ($K=0.89 \mu\text{M}$); the biochemical abstraction is dynamically confirmed, not merely static.

Limitations. Raw theme cosine and raw rank p are baseline-inflated and are NOT stand-alone evidence; this analyte's identity-specific signal is cosine +0.73 / rank separation +0.033.

guanine — purine

L1 latent 0.631 · L2 MSS 0.8745 · theme raw 0.8758 · theme identity 0.9241 · rank p 0.8727 (sep 0.0584) · top-3 0.333 · argmax nucleic_purine→nucleic_purine · Δ Purine -0.2522

Interpretation. Identity-specific preservation across the hierarchy: both the latent fingerprint and the distinctive biochemical abstraction transfer to silver.

Limitations. Raw theme cosine and raw rank p are baseline-inflated and are NOT stand-alone evidence; this analyte's identity-specific signal is cosine +0.92 / rank separation +0.058. No dynamic perturbation validation exists for this analyte (Level 4 not measured).

glucose — saccharide

L1 latent 0.2555 · L2 MSS 0.5358 · theme raw 0.8458 · theme identity -0.4605 · rank p 0.8545 (sep -0.0096) · top-3 0.667 · argmax saccharide_glycan→nucleic_purine · Δ Purine 0.024

Interpretation. Partial mid-level transfer: motif structure carries over but the identity-specific theme abstraction is weak — a surface-physics limit, not a representation error.

Limitations. Raw theme cosine and raw rank p are baseline-inflated and are NOT stand-alone evidence; this analyte's identity-specific signal is cosine -0.46 / rank separation -0.010. No dynamic perturbation validation exists for this analyte (Level 4 not measured). Weak serum-matrix recoverability (displacement 0.014): competition on colloid suppresses recovery.

uracil — pyrimidine

L1 latent 0.0562 · L2 MSS 0.4858 · theme raw 0.45 · theme identity 0.1591 · rank p 0.6455 (sep -0.0225) · top-3 0.333 · argmax nucleic_pyrimidine→nucleic_purine · Δ Purine 0.1088

Interpretation. Partial mid-level transfer: motif structure carries over but the identity-specific theme abstraction is weak — a surface-physics limit, not a representation error.

Limitations. Raw theme cosine and raw rank p are baseline-inflated and are NOT stand-alone evidence; this analyte's identity-specific signal is cosine +0.16 / rank separation -0.022. No dynamic perturbation validation exists for this analyte (Level 4 not measured).

15 · Verdict

Raman → Ag-SERS transfer is a preservation **hierarchy**, not a score. Latent coordinates transfer partially (0.42, adsorption-limited); motifs better (0.74); the broad biochemical neighbourhood best in raw terms (theme 0.92, rank 0.87) — but that is largely a compositional baseline. Corrected, identity-specific theme preservation is selective (identity 0.11, rank separation +0.01, top-3 0.67, argmax 35%), because silver homogenises most analytes toward a purine attractor ($\Delta\text{Purine} \propto -\text{adsorption fidelity}$, $p=0.006$). A minority — adenine foremost — redistribute their latent fingerprint yet retain a dose-responsive theme; that functional validation, though rare, is the strongest rung. GAIRA's job is not to maximise agreement but to separate surface physics from biochemical meaning, and to report the whole ladder.

16 · Interpretation guide

Do not say “theme preserved / failed,” “the model recovers X,” “SERS agrees with Raman.”

Say instead “identity-specific preservation is high / selective / weak,” “X's biochemical abstraction transfers / is pulled into the purine attractor,” “agreement survives to Level n of the hierarchy.” Always pair a raw metric with its null; describe failures as surface physics; treat dynamic perturbation as the gold standard — available for only three analytes.

Appendix A · Full per-analyte metrics (51 analytes)

All layers, sorted by theme rank ρ . Metrics: L1 latent · L2 MSS · L3a theme raw · L3b theme identity · L4 rank ρ · L4 rank separation · L5 top-3 · L6 argmax · Δ Purine.

analyte	family	L1	L2	L3a	L3b	rank ρ	rank Δ	top3	argmax	Δ Pur
phosphatidylinositol	lipid	0.834	0.956	0.989	0.469	0.991	0.010	1.000	True	-0.021
coenzyme a	cofactor	0.649	0.901	0.985	0.718	0.982	0.004	1.000	True	-0.023
glutathione	cofactor	0.727	0.955	0.986	0.648	0.982	0.018	1.000	True	0.042
hypoxanthine	purine	0.844	0.964	0.991	0.827	0.955	0.073	0.667	True	0.004
phosphate	organic_acid	0.361	0.649	0.918	0.603	0.955	0.010	1.000	False	0.002
glycogen	polysaccharide	0.684	0.902	0.957	0.065	0.955	-0.001	1.000	False	0.045
creatinine	small_nitrogenous	0.688	0.652	0.966	0.875	0.955	0.010	0.667	True	-0.066
valine	amino_acid	0.425	0.820	0.967	0.207	0.946	0.009	1.000	False	0.008
mannose	saccharide	0.601	0.738	0.884	-0.606	0.946	0.016	0.667	False	0.081
cysteine	amino_acid	0.580	0.809	0.961	0.694	0.946	0.017	1.000	True	-0.100
galactose	saccharide	0.446	0.797	0.924	-0.245	0.936	0.002	1.000	False	0.038
methionine	amino_acid	0.557	0.874	0.955	0.812	0.936	0.006	1.000	True	-0.103
histidine	amino_acid	0.487	0.868	0.926	-0.779	0.936	0.010	0.667	False	0.066
glutamate	amino_acid	0.334	0.655	0.954	0.153	0.927	0.020	1.000	False	0.058
xanthine	purine	0.815	0.930	0.975	0.956	0.927	0.047	1.000	True	-0.108
riboflavin	cofactor	0.339	0.639	0.972	0.709	0.918	0.011	0.667	True	-0.025
leucine	amino_acid	0.437	0.776	0.935	-0.636	0.918	-0.003	0.667	False	0.048
proline	amino_acid	0.310	0.843	0.869	-0.733	0.909	0.015	0.667	False	0.100
n-acetylglucosamine	saccharide	0.599	0.853	0.901	-0.628	0.900	0.016	0.667	False	0.105
tryptophan	amino_acid	0.339	0.679	0.930	0.214	0.882	-0.001	0.667	True	0.064
cholesterol	lipid	0.336	0.814	0.918	0.111	0.882	0.090	0.667	False	0.062
guanine	purine	0.631	0.875	0.876	0.924	0.873	0.058	0.333	True	-0.252
acetyl-coa	cofactor	0.649	0.901	0.980	0.508	0.873	-0.012	0.667	True	-0.036
isoleucine	amino_acid	0.593	0.830	0.938	-0.904	0.873	-0.007	0.667	False	0.080
phosphoenolpyruvate	organic_acid	0.618	0.821	0.933	0.867	0.873	0.073	0.667	True	-0.015
tyrosine	amino_acid	0.236	0.629	0.947	0.187	0.873	0.020	0.667	True	0.021
adenine	purine	0.359	0.741	0.947	0.733	0.864	0.033	0.667	True	-0.116
stearate	lipid	0.321	0.372	0.566	0.078	0.855	0.083	0.667	False	0.069
glucose	saccharide	0.256	0.536	0.846	-0.461	0.855	-0.010	0.667	False	0.024
ergothioneine	cofactor	0.285	0.726	0.958	0.251	0.855	-0.089	0.667	True	-0.025
alanine	amino_acid	0.255	0.662	0.887	-0.484	0.836	0.009	0.667	False	0.065
fructose-6-phosphate	saccharide	0.347	0.693	0.829	-0.730	0.836	0.017	0.667	False	0.125
lactate	organic_acid	0.402	0.565	0.782	-0.634	0.836	-0.006	0.333	False	0.121
hydroxyproline	amino_acid	0.241	0.632	0.824	-0.535	0.827	0.024	0.333	False	0.112
urate	purine	0.644	0.749	0.913	-0.566	0.827	-0.058	0.667	False	0.137
albumin	protein	0.603	0.879	0.929	-0.193	0.818	0.074	0.333	False	0.076
fructose	saccharide	0.341	0.582	0.753	-0.600	0.818	-0.000	0.667	False	0.120
citrate	organic_acid	0.618	0.696	0.843	0.852	0.809	0.007	0.667	False	-0.093
serine	amino_acid	0.354	0.690	0.850	-0.614	0.800	0.018	0.333	False	0.133

analyte	family	L1	L2	L3a	L3b	rank ρ	rank Δ	top3	argmax	ΔPur
acetoacetate	organic_acid	0.423	0.772	0.917	0.446	0.764	-0.087	0.667	False	0.038
ascorbate	organic_acid	0.392	0.674	0.861	-0.582	0.764	-0.021	0.667	False	0.080
glycine	amino_acid	0.246	0.897	0.812	-0.253	0.764	0.018	0.333	False	0.112
pyruvate	organic_acid	0.608	0.803	0.933	0.664	0.754	-0.000	0.667	True	-0.007
arginine	amino_acid	0.494	0.743	0.818	-0.932	0.754	-0.009	0.667	False	0.139
glycerol	polyol	0.470	0.596	0.804	-0.760	0.727	-0.010	0.333	False	0.137
oleate	lipid	0.264	0.365	0.507	-0.468	0.700	0.038	0.333	False	0.176
uracil	pyrimidine	0.056	0.486	0.450	0.159	0.645	-0.022	0.333	False	0.109
urea	small_nitrogenous	0.667	0.522	0.883	0.902	0.645	0.036	0.333	False	0.096
phenylalanine	amino_acid	0.301	0.530	0.831	-0.092	0.645	0.022	0.333	False	0.130
triolein	lipid	0.314	0.369	0.522	-0.357	0.636	0.026	0.333	False	0.163
thymine	pyrimidine	0.270	0.542	0.740	0.556	0.218	-0.006	0.333	True	-0.085

Appendix B · Provenance & reproducibility

Frozen atlas fingerprint: 09ed804a40836f4a05a91ba10900cded — verified unchanged. NMF (k=24), preprocessing, MSS generation, ontology, registry, and theme weights are all unchanged; SERS is projected through the fixed basis, never used to fit it.

Reproducibility vs the V2 analysis (max absolute difference): component cosine 0.0, theme raw 0.0, theme identity 0.0, MSS 0.0 — every shared metric reproduced bit-for-bit.

Sources. Level 1–3 metrics: results/v5_rebuild/representation_hierarchy_v3/ (reanalysis) and results/v5_rebuild/pure_ag_sers_theme_preservation/ (V2). Perturbation: foundation_audit/tables/validation_results.json. Matrix: spike_validation/tables/phase7_serum_vs_pure.csv. Figures reproduced from the committed, audited PNGs. Interactive versions: Foundation Explorer V1/V2/V3 (streamlit).

Scope. 51 pure analytes matched between the Gobbato Raman reference and pure Ag-SERS. This report is a document version of Foundation Explorer V3, incorporating the V1 (latent) and V2 (theme) analyses. Nothing is retrained or modified.